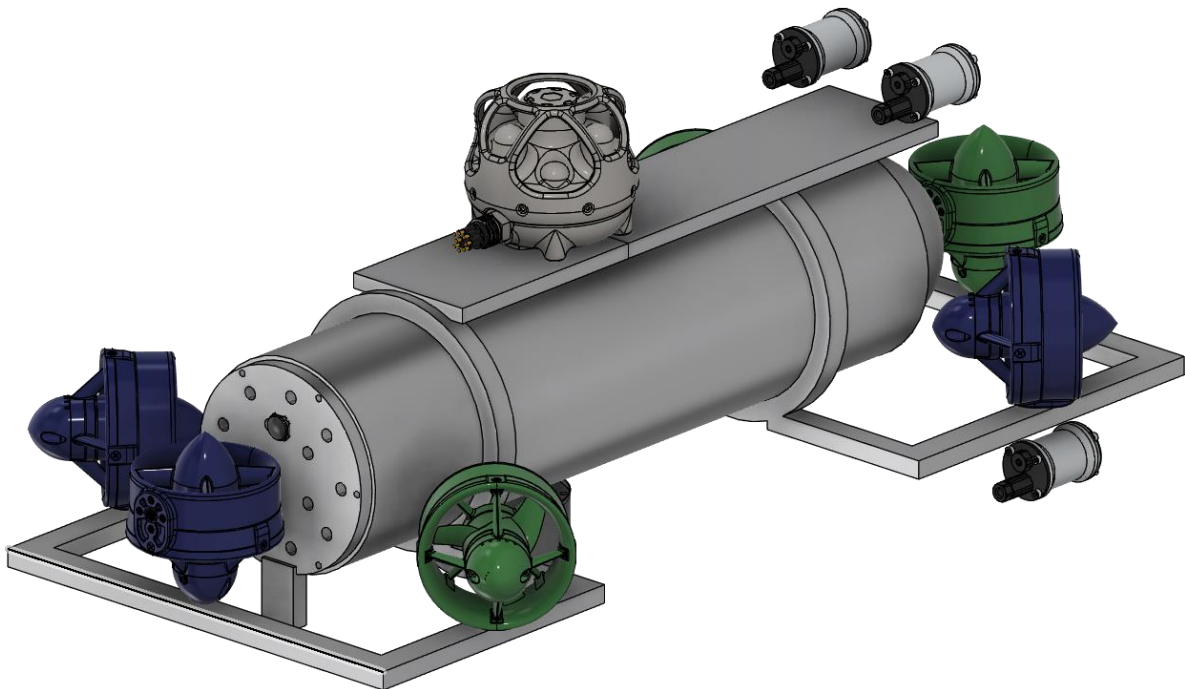


HAUV

Hybrid Autonomous Underwater Vehicle



Operator's Manual

How to set up, drive and look after the HAUV

Draft v1.0
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About the HAUV

What the HAUV is

The HAUV is a small underwater robot. You drive it from a laptop on the shore or on a boat. It can also drive itself to a point on the map.

The two computers inside

The HAUV has two computers. They do different jobs. It helps to know which one does what when something goes wrong.

Computer	What it does
UP Board	The “brain”. It thinks. It talks to your laptop, reads the GPS and the DVL, decides where to go, and runs the safety rules.
ESP32	The “muscles”. It turns the motors, switches the lights, and reads the depth sensor, the compass and the leak sensor.

The two computers talk to each other over a small cable inside the vehicle. If that link breaks, the green indicator light (Figure B, #2) goes off (see Chapter 10).

The main parts

Figure A shows the vehicle with the main parts numbered.

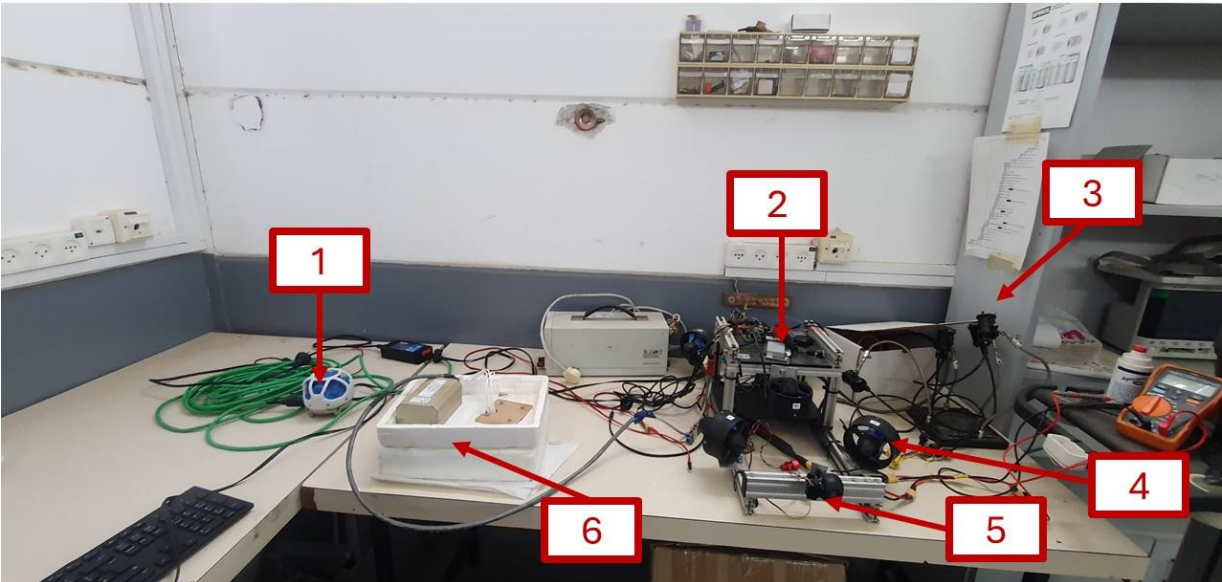


Figure A — The HAUV and its main parts. The numbers match the table below.

#	Part	What it is for
1	Subsonus (USBL modem)	Talks to the surface with sound, so the vehicle can be controlled with no cable.

#	Part	What it is for
2	Electronics tray	Holds the two computers and the boards. See Section 1.4.
3	Lights	Two sets of lamps. You switch them from the controller.
4	Thrusters	The propellers that move the vehicle.
5	Camera	Sends the live picture to your screen.
6	DVL	Measures speed over the sea floor.

Inside the vehicle — the electronics

Figure B shows the electronics tray with the main boards numbered. You do not need to touch these to drive the vehicle. It helps to know where they are when something stops working.

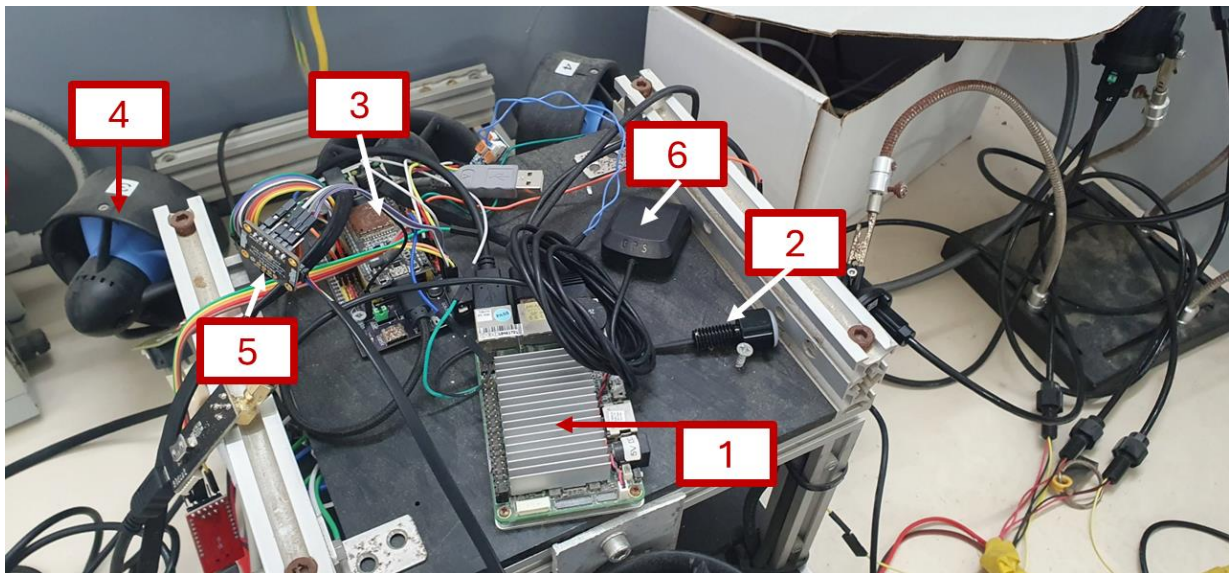


Figure B — The electronics tray. The numbers match the table below.

#	Part	What it is for
1	UP Board	The “brain” — the board with the big metal heatsink.
2	Indicator lights	2 lights, red and green next to each other. See Chapter 10.
3	ESP32	The “muscles” — the board with the coloured wires.
4	Thruster	One of the propellers.
5	I ² C connector	Goes to the temperature sensor.
6	GPS antenna	The small black square marked GPS. Gives the position on the surface.

What you need before you start

1. The HAUV vehicle, charged.
2. A laptop with **QGroundControl** installed. We call it **QGC**.

3. An Xbox-style game controller, plugged into the laptop.
4. A network cable (for the cable mode), *or* the two Subsonus modems (for the sound mode).

Safety First

Warning! The propellers can cut your fingers. They can start turning at any time when the vehicle is powered on. Keep hands, hair and clothes away.

The golden rules

1. **Never** stand next to the propellers when the vehicle is powered on.
2. **Always** power the vehicle off before you carry it.
3. Do **not** open the vehicle when it is wet. Dry the outside first.
4. Do **not** run the thrusters in air for a long time. They need water to stay cool.
5. If you see water inside, stop. Take the vehicle out. Dry it. Find the leak.
6. Do not charge a battery that is hot or damaged.
7. One person drives. One person watches the vehicle and the cable.

Before every dive

1. Look at the vehicle. Are all screws in place? Is anything broken?
2. Check that all plugs are pushed in fully.
3. Check that the lid is closed and tight.
4. Check the battery is charged.
5. Check the propellers turn freely and nothing is stuck in them.

Note. If anything looks wrong, do not dive. It is always cheaper to fix it on the table than at the bottom of the sea.

Network Setup

The laptop, the vehicle and the sensors all talk over a network. Each one has its own address. The addresses must be correct or nothing will connect.

The addresses

Device	Address	Notes
Your laptop (QGC)	192.168.168.100	You set this by hand.
UP Board (the brain)	192.168.168.101	Fixed. Do not change.

Device	Address	Notes
DVL	192.168.168.102	Fixed. Do not change.
Subsonus on the surface	168.254.1.80	Only for the sound mode.

Setting your laptop address

1. Open **Network settings** on your laptop.
2. Find the wired (Ethernet) connection.
3. Choose **Manual / Static** address.
4. Type address 192.168.168.100 and mask 255.255.255.0.
5. Save.

Tip. To test it, open a terminal and type `ping 192.168.168.101`. If you get replies, the network is good. If you get nothing, check the cable and the address.

Connecting QGroundControl

There are two ways to talk to the HAUV:

Way	When to use it
Cable (Ethernet)	Normal work. Fast. You get video and all the numbers.
Sound (Acoustic / USBL)	When the vehicle is under water with no cable. Slow. No video. Only the important numbers.

Connecting with a cable (Ethernet)

This is the normal way. Do this first when you are learning.

1. Plug the network cable between the laptop and the vehicle.
2. Power the vehicle on.
3. Open **QGroundControl** on the laptop.
4. Click the **Q** logo in the top left corner, then **Application Settings**.
5. Choose **Comm Links** on the left.
6. Click **Add**.
7. Fill in the window exactly as shown in Figure C:
 - **Name:** HAUV UP Board
 - **Type:** UDP
 - **Port:** 14550

- **Server Address:** 192.168.168.101:14550, then press **Add Server**.
- 8. Click **Save**.
- 9. Select HAUV UP Board in the list and click **Connect**.

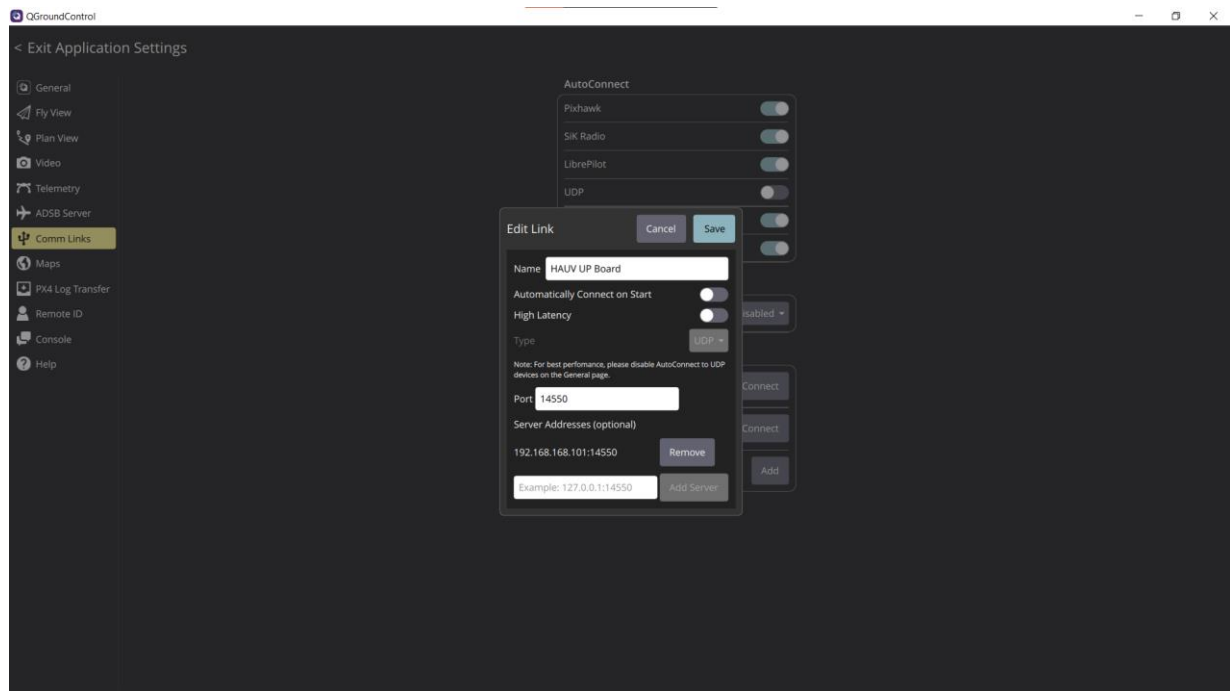


Figure C — The QGC “Edit Link” window. Name, port 14550 and the server address 192.168.168.101:14550 must match exactly.

Note. Turn **AutoConnect to UDP** off on the **General** page. If you leave it on, QGC may connect to the wrong thing and the vehicle will look dead.

When it works you will see the vehicle appear on the map and the numbers at the bottom of the screen will start moving.

Connecting with sound (Acoustic / USBL)

Use this when the vehicle is under water and there is no cable. It is slower. You will **not** get video.

You need two Subsonus modems: one on the surface (in the water, next to you) and one on the vehicle.

1. Put the surface modem in the water. It must be in the water to work.
2. Power both modems on.
3. Open the surface modem page in a web browser. Check the settings match Figure D (address 168.254.1.80).
4. On the vehicle, start the system in sound mode (see Section 5.3).
5. On the laptop, run the bridge program **once**:

```
python3 tools/acoustic_qgc_bridge.py
```
6. In QGC, connect to 127.0.0.1:14551.

The screenshot shows the 'Ethernet Configuration' page for the Subsonus surface unit. The current IP address is 168.254.1.80. The configuration fields are as follows:

Field	Value	Action
Unit Name	surface	Set To Default
Ethernet Mode	Static IP	Set To Default
Fixed IP Address	168.254.1.80	Set To Default
Netmask	255.255.255.0	Set To Default
Default Gateway	168.254.1.1	Set To Default
Broadcast Address		Set To Default
DNS Server Address	168.254.1.1	Set To Default
Link Mode	Auto Negotiation	Set To Default

Update Network Settings

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Figure D — The Subsonus surface unit settings page. The address must be 168.254.1.80.

Warning! Run only **one** copy of the bridge program. If you start it twice, the commands fight each other and the vehicle behaves strangely.

Warning! Both modems must be in the water. Out of the water they cannot hear each other and you will see **ACOUSTIC LINK LOST** on the screen.

DVL settings (for reference only)

The DVL is already set up. You do not normally touch it. Figure E shows the correct settings, in case you ever need to check them.

The screenshot shows the 'Pathfinder uDVL Network Configuration' page. The network settings are as follows:

Field	Value
Network Configuration	STATIC
Base IP Address	192.168.168.102
Network Mask	255.255.255.0
Gateway Address	192.168.168.1
Name Server Address	192.168.168.1
Command/Control Port	1033

Link Status: Not Active

Output Formats:

FORMAT	ENABLE	PROTOCOL	ADDRESS	PORT
PD0	ON	UDP	192.168.168.1	1034
PD4	OFF	UDP	192.168.168.1	1035
PD5	OFF	TCP	192.168.168.1	1036
PD6	ON	UDP	192.168.168.101	1037
PD13	OFF	TCP	192.168.168.1	1038

Submit Reset

Figure E — DVL network page. Address 192.168.168.102, command port 1033, outputs PD0 and PD6 switched ON.

Starting the HAUV

The batteries

The HAUV runs on two 12 V lithium battery packs, shown in Figure F. One pack feeds the computers, the other feeds the thrusters. The thick red and black wires join them to the vehicle with yellow plugs.



Figure F — The two 12 V battery packs and how they connect. Red is plus (+), black is minus (-). The small green board is the voltage converter that drops 12 V down for the electronics.

Connecting a battery:

1. Check the vehicle is switched **off** first.
2. Match the colours: red to red, black to black.
3. Push the yellow plug straight in until it clicks. Do not force it sideways.
4. Check nothing is loose and no bare metal is showing.

Warning! Never join red to black. A short circuit makes a big spark, ruins the battery and can start a fire.

Warning! Do not charge a battery that is hot, swollen or damaged. Do not leave batteries charging without somebody nearby.

Note. Charge both packs after every dive. A pack left flat for a long time can be damaged for good.

Power on

1. Check Chapter 2 again. Hands away from the propellers.
2. Switch the vehicle on.

3. Wait about 30 seconds. The computers need time to wake up.

Starting the software

All the software is started with one tool called `hauv.sh`. You run it on the UP Board (the vehicle's brain), through a terminal.

Command	What it does
<code>./hauv.sh start</code>	Starts everything, cable mode. This is the normal one.
<code>./hauv.sh start --acoustic</code>	Starts everything, sound mode (no cable).
<code>./hauv.sh stop</code>	Stops everything.
<code>./hauv.sh restart</code>	Stops, then starts again.
<code>./hauv.sh check</code>	Tests every sensor and says OK, LOW, FAIL or --.
<code>./hauv.sh status</code>	Shows what is running.
<code>./hauv.sh topics</code>	Lists the live data channels.
<code>./hauv.sh echo DEPTH</code>	Shows live values, for example the depth.
<code>./hauv.sh view mavlink</code>	Opens the log of one program. Press Ctrl+A then D to leave.
<code>./hauv.sh help</code>	Shows help and the short names.

Note. The first time you start the HAUV and connect QGC, it does not open on the main screen. It opens on the Vehicle Configuration screen and shows an error that parameters are missing from the firmware (see Figure G). This happens because QGC thinks the vehicle is running ArduSub. It is not a problem. Click OK, then go back to the main screen (the Fly View).

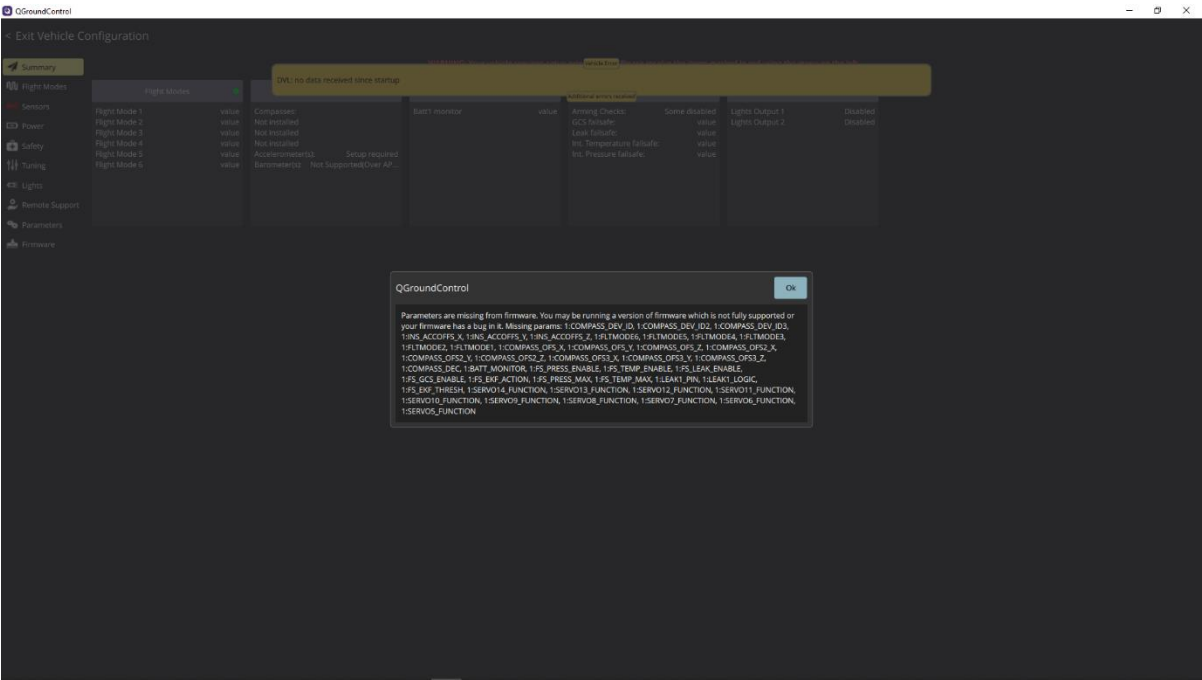


Figure G — The QGC startup screen. On first connection QGC opens here with a firmware-parameter error. This is normal — click OK to go to the main screen.

Check everything works

Before you put the vehicle in the water, run:

```
./hauv.sh check
```

It tests each sensor for a few seconds and prints **OK**, **LOW**, **FAIL** or **--**.

Result	Meaning
OK	Good. Nothing to do.
LOW	The sensor works but sends data slowly. Usually still fine.
FAIL	No data at all. Fix this before diving.
--	This sensor only sends data sometimes. Not a problem.

Note. The DVL will say **FAIL** when the vehicle is out of the water. This is normal. The DVL needs water and a sea floor to see. It will work once you are in the water.

Reading the Screen

Figure H shows the main QGC screen, called the **Fly View**. This is what you look at while driving.

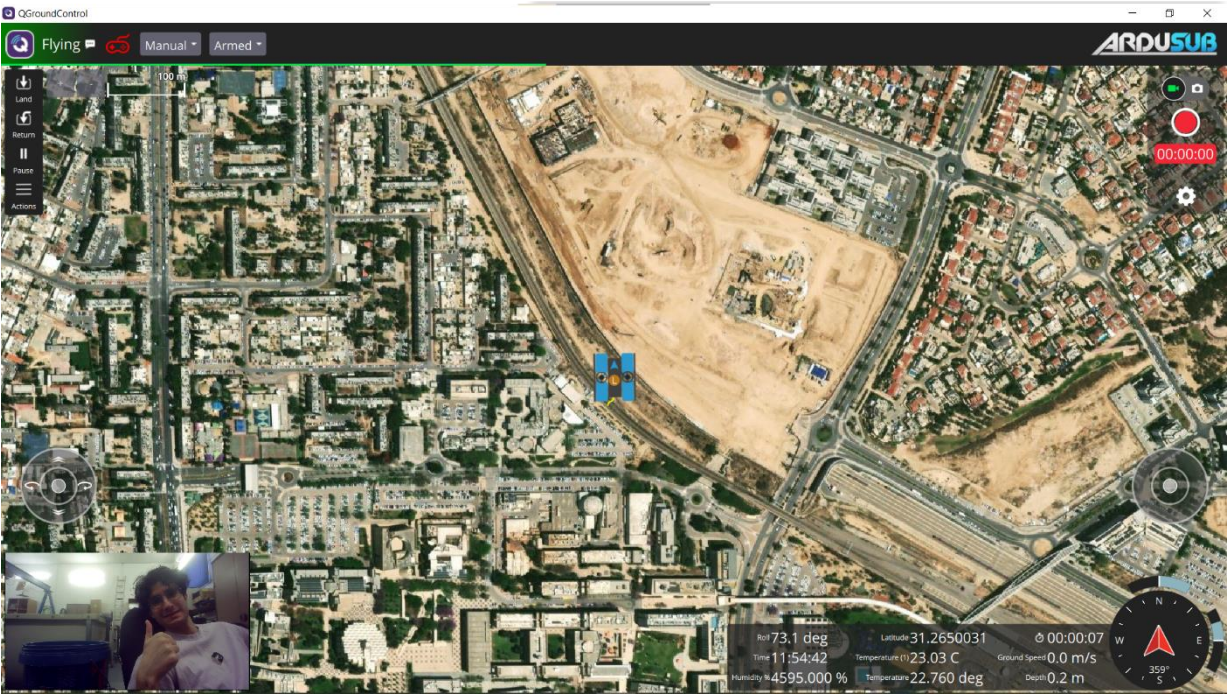


Figure H — The QGC Fly View. The yellow bar at the top is a warning message — here it says ACOUSTIC LINK LOST.

The parts of the screen

Where	What it shows
Middle	The map. The small vehicle symbol is your HAUV.
Top, yellow or red bar	Warning messages. Read these. See Section 6.3.
Bottom left	The camera picture. Says “WAITING FOR VIDEO” if there is none.
Bottom right	The numbers: depth, roll, temperature, speed, position, time.
Far bottom right	The compass. Shows which way the vehicle points.
Left side	Buttons: Land, Return, Pause, Actions.
Top right	Video record and photo buttons, and the settings gear.

The numbers you must watch

Number	Why it matters
Depth	How deep you are, in metres. Watch this all the time.
Roll	How much the vehicle is leaning. A big number means it is tipping over.
Ground Speed	How fast you are moving.
Latitude / Longitude	Where you are on the map (surface only).
Temperature	The water temperature.

Warning messages

Messages appear in a coloured bar at the top of the screen. Red and yellow ones are important.

Message	What to do
LEAK! Auto-surfacing	Most serious. Water is inside. The vehicle is already swimming up by itself. Get it out of the water. See Section 9.1 .
ACOUSTIC LINK LOST	The sound link stopped. Check both modems are in the water and powered. See Chapter 12 .
FAIL: ...	A sensor stopped working. The name after FAIL is the sensor.

Note. The leak message repeats every 3 seconds while the leak is there, so you cannot miss it.

Driving the HAUV

The controller

Hold the controller with both hands. The two sticks do the moving. The numbers in Figure I match the table below.

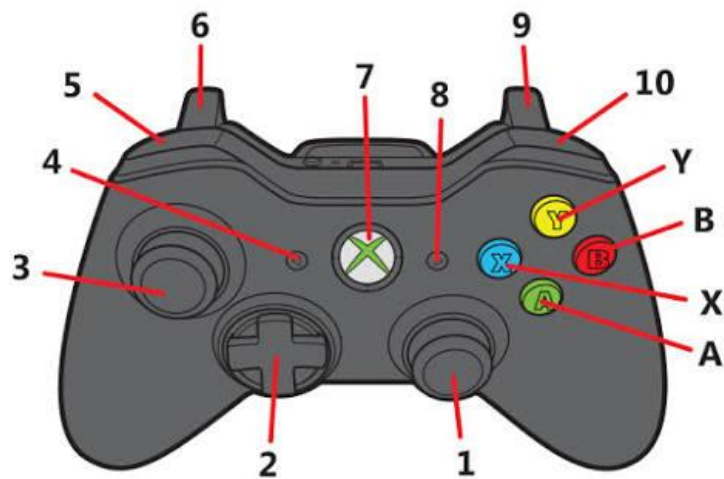


Figure I — The controller. Only the parts listed in the table do anything on the HAUV — the rest are not used.

#	Control	What it does on the HAUV
3	Left stick	Up / down: move forward and backward. Left / right: slide sideways (crab). The nose keeps pointing the same way.
1	Right stick	Left / right: turn left and right. Up / down: go up and down in the water.
A	A button	Switch one set of lights on and off.
B	B button	Switch the other set of lights on and off.
5	LB shoulder	Tilt the camera up.
10	RB shoulder	Tilt the camera down.

The parts **not** used for driving are: 2 (D-pad), 4 (Back), 6 and 9 (the two triggers), 7 (Xbox button), 8 (Start), and the X and Y buttons.

Tip. Push the sticks gently. Small movements are enough. The vehicle is heavy and keeps moving after you let go.

Your first drive

1. Put the vehicle in the water. Hold it until you are ready.
2. Check the screen. The depth should be near 0.

3. Push the left stick forward a little. The vehicle should move forward.
4. Pull it back. The vehicle should stop and go back.
5. Turn with the right stick. Watch the compass move.
6. Push the right stick down a little to go deeper. Watch the depth number grow.

Warning! Always know where the vehicle is. If you cannot see it and the screen is frozen, stop and bring it back.

Flight Modes

The vehicle has three modes. You change them in QGC, at the top of the screen.

Manual

You control everything with the controller. The vehicle does only what you tell it. Use this most of the time.

Stabilize

The vehicle tries to stay level by itself. You still drive it. Use this if the vehicle is rolling or tipping too much.

Guided — “Go to location”

The vehicle drives itself to a point you choose on the map. It keeps its depth on the way, and when it arrives it stays there instead of drifting away.

How to use it:

1. Make sure the vehicle has a GPS position (it must be on the surface).
2. On the map, click the place you want to go.
3. Choose **Go to location**.
4. The vehicle turns towards the point, then swims to it.
5. When it arrives, it stays there and holds its position and depth.

How to stop it (any of these work):

1. Move any stick on the controller.
2. Change the mode to **Manual** in QGC.
3. If the link is lost for 5 seconds, the vehicle stops by itself.

Note. While going to the point, the vehicle keeps the depth it had when you pressed **Go to location**. If you were on the surface, it stays on the surface.

Safety Features

These work by themselves. You do not switch them on.

Leak — the vehicle swims up by itself

If water gets inside, the leak sensor sees it. Then:

1. The vehicle **stops** all sideways movement.
2. It drives its up/down thrusters to **go up** to the surface.
3. It stays on the surface.
4. Your screen shows **LEAK! Auto-surfacing** in red, again and again.

This ignores everything else. Even if you are driving, or the vehicle is going to a point by itself, the leak wins and the vehicle goes up.

What you must do:

1. Do not fight it. Let it come up.
2. Take the vehicle out of the water.
3. Dry the outside. Then open it and find the water.
4. Do not dive again until you find why water came in.

Note. This works even if the cable or the sound link is broken. The decision is made inside the vehicle, not on your laptop.

Note. When the sensor is dry again for 5 seconds, the vehicle gives control back to you automatically. If the sensor stays wet, it stays on the surface — that is on purpose.

Lost control link

If the vehicle stops hearing your controller for about 1.5 seconds, it sets all thrusters to stop. It will not keep driving away with the last command.

Indicator Lights

There is a **green** light indicator. It tells you if the two computers are talking to each other. You can see where it sits in Figure B — it is number 2.

Green light	Meaning
On, steady	Good. The two computers are talking. Sensors and motors work.
Off	Bad. They are not talking. The motors will not answer. See Chapter 12.

Note. Just after you switch the vehicle on, the green light is on for a short time before the connection is really made. Wait about 30 seconds before you trust it.

A **red** light indicator sits right next to the green one and shows leaks, faults, and when the propellers are live — keep clear of the propellers whenever it is on (see Chapter 2).

Red light	Meaning
Fast Blink	Leak Detected
Solid	Sensor Fault
Slow Blink	Props Live – Keep Clear
Off	All Clear

After the Dive

1. Bring the vehicle out of the water. Two people if it is heavy.
2. Switch the vehicle off.
3. Wash the outside with **fresh** water. Salt water eats metal.
4. Pay attention to the propellers. Remove weed, sand and plastic.
5. Dry the vehicle with a cloth.
6. Only when it is dry outside, open it and check for water inside.
7. Charge the battery.

Warning! Never put the vehicle away wet or with a flat battery.

Troubleshooting

Common problems

Problem	What to try
QGC does not connect	1. Check the network cable. 2. Check your laptop address is 192.168.168.100. 3. ping 192.168.168.101. 4. Check the link settings match Figure C. 5. Turn AutoConnect UDP off in QGC.
Vehicle shows on screen but nothing moves	The two computers are not talking. Check the green light (Chapter 10). Restart with <code>./hauv.sh restart</code> .
No video picture	Video only works over the cable. Check the cable mode is running.

Problem	What to try
DVL says FAIL	Normal out of the water. If it still fails in the water, check the DVL settings in Figure E.
ACOUSTIC LINK LOST	1. Both modems must be <i>in the water</i> . 2. Check both are powered. 3. Check only <i>one</i> bridge program is running on the laptop. 4. Check the surface modem address (Figure D).
Some sensors say FAIL	Run <code>./hauv.sh check</code> to see which. Restart with <code>./hauv.sh restart</code> . If it still fails, the sensor or its plug is bad.
Vehicle drifts away in Go to location	The GPS is only accurate to a few metres. This is normal. Take manual control if you need to be exact.
Vehicle keeps swimming up and will not stop	This is the leak failsafe. There is water inside. See Section 9.1. Get it out of the water.

Useful checks

Command	Use it to
<code>./hauv.sh check</code>	See which sensors are alive.
<code>./hauv.sh status</code>	See which programs are running.
<code>./hauv.sh echo DEPTH</code>	Watch the depth value live.
<code>./hauv.sh echo LEAK</code>	Check the leak sensor (0 = dry, 1 = water).
<code>./hauv.sh view guidance</code>	Read the messages from the driving program.
<code>./hauv.sh restart</code>	The general “switch it off and on again”.

Tip. If you are lost, `./hauv.sh restart` fixes most software problems. It does not fix broken cables or wet electronics.